Transhuman Communication: Human Augmentation Technologies through Co-Speculation Workshops

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The advancement of human-computer integration as a research field promises to introduce transhumanistic ways of communicating through the enhanced abilities of augmented humans. Our work seeks to illuminate this experiential landscape, exploring a diverse set of human augmentation technologies (HATs) for communication purposes. We investigated this topic through four co-speculation workshops focusing on physical, cognitive, sensory and emotional augmentations with 35 participants. Through a reflexive thematic analysis of the workshop data, we outlined eight HAT speculations for transhuman communication, grouped into four overarching augmentation clusters: (1) Bodily Changes in/for Communication, (2) Communication with Transferrable and Collective Beings, (3) Communication through Emotion and Memory, and (4) Communication with Augmented/Altered Perception. By serving as a foundation for discussions on transhumanism and communication in CSCW and beyond, these speculations contribute to a design space highlighting design opportunities and challenges to developing and researching near-future communication technologies.

CCS Concepts: • **Human-centered computing → Human computer interaction (HCI)**; **Collaborative and social computing**.

Additional Key Words and Phrases: Transhumanism, cyborg, human augmentation, human-technology integration, wearables, speculative design, communication

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1 Introduction

Communication, as an integral part of human existence, has evolved throughout time, responding to the evaluative needs for cooperation and coordination of human and pre-human animals. The complex means of communication soon evolved through the development of natural languages, painting, sculpture, architecture, writing systems, smoke signals, and more. Beyond its many roles (related to social cooperation, preservation of memory, dissemination of knowledge, and so forth) and functions (see [51]), one key aspect of communication is its experiential dimension - where

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various modalities such as written texts, images, video, telecommunications, virtual reality, and face-to-face interactions offer unique and diverse communication experiences. After many "revolutions" in communication from the invention of alphabets to the introduction of digital networks [4], a new paradigm is appearing on the horizon, rooted in the domain of human-computer interaction: Transhumanism. Fundamentally, the paradigm suggests that the integration of machines and computers into the human body will augment individuals physically, sensorily, cognitively, and emotionally [10].

The dominant understanding of transhumanism indicates that the augmentation of humans will open up revolutionary means of communication, especially emphasizing the unification, immediacy, and efficiency of information exchange. These approaches hold many potential risks for future communication, as such a depiction of the future that endangers plurality and diversity, in favor of homogenous "perfected" transhumans [49, 69]. Additionally, numerous communication theorists insist that the imperfection and plurality of human communication have great value in enriching cultures by supporting a multiplication of perspectives and semiotic materials [38, 61]. To navigate transhuman futures in communication, as HCI and CSCW communities, we must not be limited to the ideology of transhumanism and oversimplifications on the workings of communication, and instead explore how human digital bodily augmentation technologies (HATs) can complement the already complex, diverse and imprecise nature of human communication.

To broaden this ideological view when exploring transhuman communication, in this work, we turn to feminist scholar Donna Haraway, and view transhumanism as a way to challenge traditional notions of identity and embodiment as "cyborgs"[41]. From this perspective, we argue that HATs hold immense potential to profoundly reshape and diversify future communication. Indeed, the prospect of human-computer integration [87] has led to the exploration of the affordances of transhumanism for diversifying communication experiences (i.e., social play with implantable and prosthetic technologies [76], and expressive [14, 106] and social uses [107] of artificial limbs). However, these examples only represent individual cases illustrating what is possible within the limits of current technology. Understanding the communicative landscape in transhuman societies and also the opportunities granted by the integration of technology and bodies requires a broader critical inspection.

To foster a broader conversation on transhuman communication, we ask *"How can HATs be designed to diversify communication experiences?"* Here, we refer to the word "diversify" to highlight our goal to explore the various ways augmented abilities could enable transhumans to express themselves and make meaning in communication, rather than aiming for one standardized and efficient method as implied by the mainstream view of transhumanism [49, 69]. To explore this landscape, we conducted four co-speculative workshops [16] involving 35 participants with diverse experiences in creating fiction, designing, and developing technologies. These workshops focused on creating and discussing design speculations [26], conceptual designs, and fictional scenarios around them to critically engage with how HATs could shape our communication experiences and societies. Each workshop focused on different types of augmentation, such as cognitive, physical, emotional and sensorial [10], and encouraged participants to use these as starting points to explore the future of communication. By thematically and reflexively analyzing the workshop data [12], we present 8 speculative HATs (Fig. 4) categorized into four clusters of (1) Bodily Temporalities in/for Communication, (2) Communication with Transferrable and Collective Beings, (3) Communication Through Emotion and Memory, and (4) Communication with Augmented/Altered Perception. These HATs reveal a design space with a wide set of possibilities and challenges for developing and researching transhuman communication technologies. By also discussing how HATs can currently be researched by crafting state-of-the-art technologies, our work offers concrete starting points and inspiration for researching the near future of communication in CSCW and beyond.

2 Background and Related Work

2.1 (Human) Communication

While increasingly intertwined, human and digital communication are radically different. Human communication, rather than being a mechanic process of humans reacting with uniform responses to information received - as proposed by the hypodermic needle model [7] and Wittgenstein's idea of language [103]- is a complex phenomenon where diverging interpretations are core concepts. Early communication models, such as Jakobson's functions of language [51], already confirmed this complexity by highlighting the semantic material transmitted as being far from stable. It is rather the result of a diverse negotiation between the different components of any communicative event such as a message, sender, receiver, channel, code and context. Here, the whole area of interpretative semiotics as inaugurated by C. S. Pierce [78] has focused its attention on the role of the receiver in interpreting the meaning of the texts they encounter. Texts are, according to Eco [29], lazy machines that convey only minimal amounts of information, and the meaning is then determined collectively, through an interaction between the communicative intention of the author, the interpretative choices of the reader, and the structural properties of the text [28]. Additionally, the influence of different bodies in semiotic activity is highlighted by Kristeva's concept of Chora [58]. McLuhan's notion that "the medium is the message" further underscores the impact of communication technologies on interpretation [68]. Considering these factors, designing communication technologies cannot be simplified as merely enhancing information exchange efficiency.

Investigations on the semiotics of culture suggest technology can also go in the opposite direction, in order to increase semiotic diversity. Lotman [61] claims that the stereoscopic quality of diverse interpretations of reality is strongly beneficial for culture, as it allows a sort of cultural biodiversity. Culture then produces texts that are purposefully of difficult and ambiguous interpretation (such as artworks, [62]), that allow multiple re-combinations (as games do, [95]), and practices that require continuous recordings (such as forms of auto-communication, [63]) to increase the internal indeterminacy of the system and increase the semiotic diversity of a semiosphere. With the increasing complexity of digital media, HCI and CSCW in particular unsurprisingly started to focus on how to increase semiotic diversity. For instance, the Internet expanded the possibilities for remote personal communication by providing modal opportunities for communication through emojis [45] and GIFs [86]. Social wearables [24], on the other hand, can enable communication through newly situated modes of communication such as changing the colors of garments [55]. Researchers also started to expand the non-verbal channels of communication by distributing scents during physical interactions [2, 52]. Furthermore, technology-mediated communication in VR environments provided humans with new ways to express themselves, i.e., avatars [35] and 3D visualizations of biometric data [6], exceeding the limitations of the physical world. Opportunities for real-time connections with remote individuals through a variety of means such as body-worn and VR-based live streaming [80, 104] also started to appear, giving individuals immersive modalities through which to share their perspectives during communication.

However, parts of digital communication and some reductionist approaches to cognitive sciences have brought about a resurgence of behavioral approaches to communication. In an attempt to explain large phenomena and big data, concepts such as "virality", which are little more than folk concepts with very limited heuristic potential [66], go back to the idea of a text that makes it spread "like a virus", relegating all aspects relative to contexts, media, and interpretation unjustly to the background. Dawkin's theory of "memetics" [25] goes exactly in this direction, and it follows a biological metaphor to propose a model of communication that sees it as an inevitable, mechanical fact. To face the possible (and impossible) transhuman futures of communication, we need to extend this mechanic understanding of communication by focusing on how human technological extensions can affect a system that is already complex, diverse, multimodal and blissfully imprecise, as is human communication.

2.2 Transhumanism

Transhumanism is a philosophical and technological movement that seeks to enhance the human condition through the integration of technology and biology [10]. More describes this undertaking as "[...] applying technology to ourselves, [so] we can become something no longer accurately described as human" [74]. Transhumanists advocate for life extension, cognitive enhancement, and the radical morphological freedom for people to modify their bodies with physical, cognitive, emotional and sensory augmentations as they see fit [10]. They claim that transcending human limitations and achieving immortality has been a common theme throughout history, spanning from ancient mythology to modern-day technology [9]. The advent of modern technology has already made transcending human limitations a practical possibility, and people use prosthetic limbs to enhance their physical speed [90], automated insulin delivery systems to influence their endocrine systems [17], or pacemakers to streamline their heartbeat [8]. Similarly, efforts in developing brain implants have started to pave the way for cognitively connecting to and controlling computers [37, 102]. Medical innovations like gene therapy promise a rewriting of human capabilities on a fundamental biological level, from engineering high endurance in people to achieve greater successes in sports [69] to redefining beauty through people tailoring how their body looks to the finest detail [98].

However, the perceived legacy of transhumanist thought as the "natural" result of linear "enlightened" and "rational" progression has been problematized [60, 81]. As designers, developers and researchers, we need to be aware that technologies do not just appear out of thin air: They are developed by people with their own experiences, values and worldviews, and in this way, technological progress is socially constructed and negotiated [21]. Here, transhumanism is faced with a "values problem" [81], and as described by Porter: If we pursue being "better", "healthier" or "happier" through HATs as the foundation of going beyond current human limitations, we are necessarily forced to define these attributes in relation to the status quo and currently existing norms [81]. The normative tendencies of technology have already begun to be problematized as reductionist and essentialist currents within HCI and beyond: Spiel et al. revealed that the technologies for people with ADHD are primarily utilized to manage behaviors considered as disruptive according to neurotypical standards [92]. Spors et al.'s examination of self-care apps for mental health reveals that these apps tend to oversimplify mental health, presenting it as an individualistic endeavor rather than a complex and relational undertaking [93]. Many, if not all, of the papers and products critiqued in the afore-mentioned research are motivated by "improving" people's lives and their health. Without humility, nuance and care, this leads to the promotion of "dominant paternalistic paradigms" [92]. Therefore, embracing transhumanist futures requires a critical awareness of the destructive potential of HATs: We have to consider the historical and ongoing harm, violence, and death caused by eugenics as a means to brutally enforce a specific genetic blueprint or phenotype in the pursuit of so-called societal "improvement" and the survival of a select few [54, 82, 84, 85]. Furthermore, transhumanism may introduce new forms of oppression, as the division between "enhanced" and "not enhanced" individuals could lead to discriminatory practices, jeopardizing the plurality and diversity of human beings in favor of a homogenous ideal [1, 49, 50, 57, 69].

Here, it becomes clear that we have to actively question the status quo to anticipate the future, otherwise, we are always at risk of creating communication interventions from a "techno-solutionist" approach that reinforces already existing norms and oppression, by focusing only on solving a perceived "problem" in the easiest way possible [71, 75].

2.3 Towards Transhuman Communication

Despite the mainstream understanding of transhumanism described in the previous section, there are alternative framings that are suitable to examine pluralistic futures for communication. Here, Haraway [41] argues that humankind actively blurs the line between human and machine, and as a result, we are already challenging traditional notions of identity and embodiment as “cyborgs”. Within this understanding, HATs have the potential to radically reconfigure how we live, work, experience the world and relate to each other, and other life forms.

Designers, researchers and developers have also begun to explore the affordances of transhumanism, and of integrating technology within our bodies/minds. Examples of such work include Yamamura et al., who explore social experiences around a robotic limb system “JIZAI ARMS”, allowing people to operate each other’s limbs, and share and exchange arms [107]. Mueller et al. developed a set of games to enhance social play through artificial limbs and implementable technology [76], while Xie et al. trialed an "augmented tail" device that offers new possibilities for gestural communication [106]. Similarly working within a playful context, Buruk et al. investigate bodily extensions for communication, highlighting unique channels for expression through modalities like light, movement and sound [14]. Beyond human-to-human communication, designers and researchers are extending their own perceptions and more-than-human ways of understanding the world, and examples include Harbisson [42], who is an augmented activist-artist and is able to perceive non-visible colors.

Several studies within the HCI field have also employed speculative design (SD) to investigate transhumanism. Thibault et al. [96] explored the concept of trans-urbanism, combining transhumanism and smart cities through fictional scenarios that assess the implications for urban design. Buruk et al. [15] presented speculative abstracts envisioning children’s technologies in the era of transhumanism, analyzing the ethical and social consequences of enhancing children’s abilities. Incorporating Afrofuturism, Bray and Harrington [13] aimed to create inclusive futures by involving marginalized communities’ experiences in the SD. Similarly, Harrington et al. [43] expanded the SD canon to include a Black future, advocating for more diverse and inclusive futures aligned with the aspirations of marginalized communities. These studies collectively demonstrate how SD can shift the focus from techno-solutionist transhuman technology development towards exploring opportunities to shape everyday experiences.

The aforementioned approaches illustrate how HATs can shape new communication experiences, and open up new design spaces. Nevertheless, although introducing concrete applications of such communication scenarios, applied augmentation instances depict individual applications showcasing current technological capabilities. On the other hand, speculative approaches, while providing provocative opportunities for transhumanism, lack focus on communication experiences. In order to comprehend the communicative terrain in transhuman societies and grasp the potentials afforded by the fusion of technology and bodies, a more comprehensive and critical examination is necessary. Accordingly, we need a careful investigation of the social, relational and experiential aspects of transhuman communication wherein we have to actively shape and guide which technological futures (plural) we collectively wish to manifest.

3 Method: Co-Speculation Workshops

Our work focuses on speculative design (SD) [26] to understand HAT in transhuman communication experiences. In HCI and CSCW, speculative design is a practice used to challenge current norms by creating designs that represent diverse societal contexts and ideas [47, 105]. By presenting speculative designs, we aim to start a conversation about the opportunities for designing transhuman communication technologies, and also the challenges they may bring to our communication experiences and society as a whole.

To utilize SD for exploring transhuman communication experiences, in this work, we conducted a series of co-speculation workshops. Co-speculation is a participatory approach to SD, aiming to involve different stakeholders in the creation of future concepts [100]. One way to foster participation in speculation is by conducting a series of co-speculation workshops [16], in which participants engage with the different facets of a design topic in individual themed workshops. In this study, we conducted four thematic workshops to engage participants with different types of transhuman augmentations (physical, sensorial, emotional and cognitive) as starting points for exploring transhuman communication.

3.1 Epistemology

The findings, insights and takeaways we discuss in this paper were made only possible through closely working with other people—fellow researchers within our team, and most importantly, the participants who speculated with us. We understand the knowledge created through these processes as 'phenomenologically situated', as being constructed out of people's own lived experiences, and being embedded within personal, cultural and societal discourses. Here, "situatedness", as conceptualized by Haraway, extends and includes science as a human activity in itself, as a negotiated, pluralistic process [40]. As a result of this "relational" epistemological stance [39], we position ourselves as being closer to the "third paradigm" in HCI, as outlined by Harrison et al. [44], and our research also leans on the concept of "entanglement HCI". This view advocates for acknowledging the complex interdependencies between humans, technologies and environments, urging designers to consider the intricate relationships within socio-technical systems [34].

3.2 Statement of Positionality

Given our inherent influence within this research, we seek to transparently articulate our values underpinning this research. We begin by outlining a statement of positionality for researchers, enabling readers to contextualize the paper's contribution. The authors of the presented study include five men and a non-binary person and their ages range from early to mid-thirties. They are based in Finland, yet their countries of origin vary from Türkiye, Finland, Italy, Germany and USA. They include researchers with backgrounds in HCI with a focus on design research (1st, 2nd and 3rd authors - all having experience in SD), and drawing from feminist, queer and disability studies (2nd author), game studies (5th and 6th authors), and semiotics (4th author). While some authors have personal experiences with body modifications (i.e. tattoos and piercings), and wearable technologies (i.e., developing and using gameful and fashionable wearables), others have focused more on low-tech approaches and medical treatments (i.e., treating asthma). The authors hold a variety of opinions regarding transhumanism, ranging from skepticism about its potential societal risks, to excitement about the playful and joyful possibilities it presents. Overall, the collective aim of the authors is to explore the intersection of technology and humans, considering the implications and possibilities of transhuman technologies while being mindful of ethical concerns and potential social impact.

3.3 Procedure

This study adhered to ethical research principles in accordance with Finland's policies. Participants were provided with a project information sheet, a privacy and data notice, a consent form, and a code of conduct for the workshops. Participants were reminded of their right to leave the workshop at any time without explanation, and facilitators emphasized the importance of open-mindedness and mutual respect. Figure 1 summarizes the activities and schedule of the workshops. The workshops

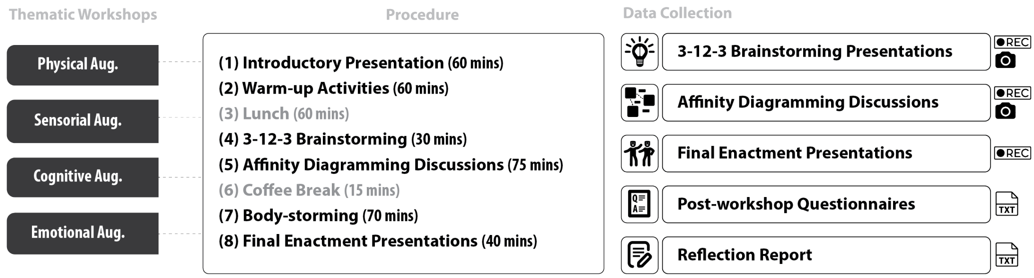

Fig. 1. The schedule, activities and data collection from workshops

started with participants signing the consent form, and their structure was similar but with slight differences:

- We began with an ***introduction*** to outline the workshop's scope for participants. Moderators explained communication channels using examples like speech, gestures, and clothing. They also discussed the influence of technology on communication, citing digital platforms like texting and social media. Transhumanism was explored through examples such as body modifications, implants (such as sensors), and prosthetics, along with concepts like brain-to-brain interfaces and medical advancements. Following this, we presented the research problem: "Transhuman technologies are coming on the horizon, yet, we don't know how they might designed for communicative purposes and have an impact on our communication experiences". This was followed by the introduction of the workshop goal, which depending on the theme was, "to brainstorm and discuss new human abilities—sensory, physical, cognitive, or emotional—as avenues for communication".
- ***Warm-up activities*** were crafted based on relevant literature and introduced to enhance creativity and divergent thinking [3] (e.g., generating multiple uses for a bottle within a time limit), promote mindfulness and body awareness [53] for role-playing activities (e.g., a short meditation followed by a game of tag acting as zombies), and foster group cohesion within groups [46] (e.g., positive and affirming conversation sessions). These marked the end of the first half of the workshops, after which participants had *lunch*.
- In ***3-12-3 Brainstorming*** [36], participants take 3 minutes to generate keywords, 12 minutes to combine them, and then present their ideas in 3 minutes. This activity started with a generative question "What kind of new sensory, physical, cognitive or emotional abilities can be designed and utilized for communicative purposes 100 years ahead of our time?" and allowed for the quick expression of various ideas using Post-its.
- ***An affinity diagramming*** [64] session was based on furthering, discussing and grouping the ideas from the previous step with the involvement of all participants (see Fig.3-a). Moderators also encouraged participants to speculate on the communication experiences the discussed ideas might lead to.
- ***Bodystorming***[67] included paper prototyping [91]. They used scrap materials (i.e. paper, beads, textiles) to ideate HATs and prepared communication scenarios.
- In ***Final Enactment Presentations***, refined ideas were presented by demonstrating a scenario via role-playing (see Fig.3) and were discussed.

Three of the workshops were moderated by the first two authors, and one workshop was moderated only by the first author. The authors ensured the creation of diverse ideas by giving a wide range of examples in the first presentation, and encouraging participants to explore alternative

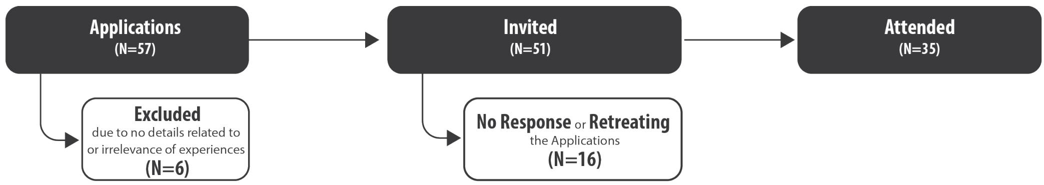

Fig. 2. A breakdown of the participant selection process.

ideas that differed from others during bodystorming and brainstorming sessions. For the latter sessions, the workshop moderators used their tacit knowledge experience [23] from previous workshops and their explicit knowledge of the topic to assess the variety of ideas. Moreover, the bodystorming and enactment activities aimed at sensitizing participants to the complex realms of transhumanism and communication through role-playing activities [99]. Each workshop lasted approximately 7 hours.

3.4 Participants

The workshops were conducted in the scope of a special topic course themed "Workshop Series on the SD on Transhuman Communication Technologies" offered at Tampere University (Finland), and were open to all students and to people from outside the university. The call was distributed through social media channels, internal communication channels of the university, and individually to authors' relevant contacts. Interested people needed to fill in a pre-application form which involved questions regarding their education, expertise in general, their experience in the creation of fiction, designing and developing communication technologies, their knowledge about transhumanism, and finally the reason why they were interested in participating.

In total, 57 people applied for the workshop and the authors selected participants by primarily considering the relevance of the participants' expertise and experience, resulting in the exclusion of 6 applications due to no relevant experiences being mentioned. We invited 51 participants (21 women, 2 preferred not to say and 28 men), yet 16 applicants either retreated their application or did not respond to our invitation (Fig.2). In the end, 35 participants (11 women, 23 men and 1 preferred not to say) attended the workshop. One participant attended from outside the university and was studying for a bachelor's degree in electronics and communication. The rest were a mixture of bachelor's (N=1), master's (N=26), and Ph.D. (N=6) students, as well as research staff (N=1) at the Tampere University. The bachelor's student was studying Computer Science (N=1). The master's students were from Master's programs in Computing Sciences (N=10/26), Information Technology (N=11/26), Sustainable Digital Life (N=2/26), Photonics Technologies (N=1/26), Teaching, Learning and Media Education (N=1/26), and Automation Engineering (N=1/26). Ph.D. students were from doctoral programs in Engineering Sciences (N=2/6), Humans and Technologies (N=3/6) and Computing and Electrical Engineering (N=1/6).

Overall, participants were selected based on their overlapping experiences in technology design/development and fiction creation, while we also considered their experiences and/or interests that might be relevant to different communication contexts. Our participation pool included participants with *design, development and research experiences* about body-oriented technologies (i.e., wearables, embodied interaction), and with technologies frequently associated with transhumanism such as robotics, virtual reality (VR), brain-computer interfaces (BCI), artificial intelligence (AI), and machine learning. These experiences helped participants identify design opportunities by speculating over the same or similar technologies for transhuman communication. Moreover, the participants also demonstrated a contextual knowledge around developing these technologies (i.e.,

Table 1. Examples of participant experiences organized based on the workshop themes and high-level categories of the experiences.

Workshop Theme	Category of Experiences	Examples from Participant Experiences
Sensorial Aug.	Design & Development	***Wearables*** *(e.g., haptics),* ***Brain-Computer Interface (BCI)****,* ***Incident Management App****,* ***Cybersecurity****,* ***AI & Machine L.****,* ***Robotics****,* ***Design Analysis****, and* ***Speculative Design*** *(e.g., playing in dreams)*
	Specific to Communication	***Language Teaching*** *and* ***Experience with ALS***
	Fiction Creation	***Role Playing Games*** *(e.g., players and development) and* ***Amateur Fiction*** *(e.g., creative writing and short stories)*
Physical Aug.	Design & Development	***Wearables*** *(e.g., smart helmets and gesture detection),* ***Robotics*** *(e.g., for disabled patients, and non-verbal communication with robots),* ***Embodied Interaction*** *(e.g., rehabilitation of children with ADHD),* ***AI & Machine L.****, and* ***VR***
	Specific to Communication	***Language Teaching*** *(game-based learning)*
	Fiction Creation	***Role Playing Games*** *(e.g., players and development) and* ***Amateur Fiction*** *(e.g., short stories and animation)*
Emotional Aug.	Design & Development	***Wearables & BCI*** *(e.g., a textile-based EEG for detecting mood),* ***Human-AI Interaction & Robotics*** *(e.g., emotion-driven human-collaborative robot interaction, a waiter robot),* ***Service Design*** *(e.g., redesign of Tinder as an inclusive platform for LGBTQ), and* ***UX Design & Usability Testing***
	Specific to Communication	***Television, Film and Photography***
	Fiction Creation	***Role Playing Games*** *(e.g., players), and* ***Amateur Fiction*** *(e.g., short stories, movies and theater plays)*
	Psychology	***Human Emotions & Cognition***
Cognitive Aug.	Design & Development	***Wearables & IoT*** *(e.g., wearables for monitoring and tracking cognitive functions, and detecting driver's sleepiness by heart rate, eye condition and stress level),* ***AI & Machine Learning*** *(e.g., human-centered AI, brain-inspired machine learning),* ***UX & Interaction Design****, and* ***Product Design***
	Specific to Communication	***Information Security***
	Fiction Creation	***Role Playing Games*** *(e.g., moderators and Players), and* ***Amateur Fiction*** *(e.g., short stories)*

cybersecurity, non-verbal and emotion-driven communication - with robots-, mood detection, as well as designing them for diverse populations such as children with ADHD, LGBTQ, and disabled

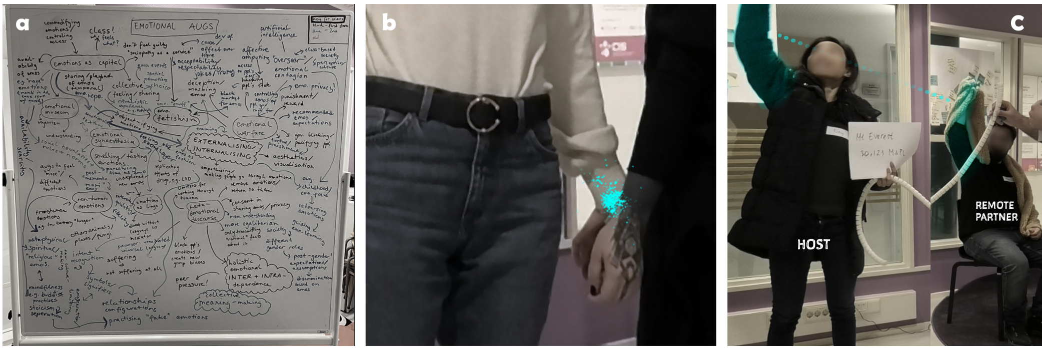

Fig. 3. Instances from the workshops: (a) affinity diagramming board from the emotional augmentation workshop, (b) the enactment of transhumans exchanging memory in the sensory augmentation workshop, and (c) temporal connection between remote partners in the cognitive augmentation workshop.

people) which was crucial for speculating on the impact of augmentations in communication experiences. Also related to this, some of the selected participants had *specific communication-related experiences* (i.e., teaching a foreign language, having a relative with ALS disease, studying media, or information security) which contributed to enriching ideas by providing relevant communication contexts. Finally, the *fiction creation* experiences of participants varied from role-playing games to creating amateur fiction such as short stories, movies and theater plays. These helped with the creation of fictional worlds and characters for transhuman communication. Examples from these experiences are summarised in Table 1.

Considering the participants' experiences, we categorized and communicated roles to them during the workshops, including *Designers & Developers* (N=14), *Fiction Creators* (N=13) and *Others* (N=8). The last category consisted of diverse backgrounds that would not fit the previous categories. Although we allocated the role of "Communicators" to these individuals, we note that we see communication as inherent to all human lives, and that everyone experiences communication-related issues (such as desires to express themselves, concerns regarding their privacy, and so on). In this regard, all of the participants acted as experienced practitioners of daily communication, bringing *"their body, future dreams, fears, and condition"* [77] when creating speculations and providing contexts, desires and values for transhuman communication.

Each workshop included 8-10 participants working in groups of 2-4 people each. In each workshop, groups included at least one fiction creator and one participant from the developers and designers category. Participants who were enrolled at Tampere University received three study credits for taking part, and non-students (N=1) received a 50 Euro shopping voucher.

3.5 Data Collection

During the workshops, data was captured through photo, video, and audio recordings. Photos were taken of 3-12-3 ideation results (Post-its) and affinity diagramming boards (see Fig.3-a). Video and audio were recorded during the group presentations of 3-12-3 ideas, affinity diagramming discussions, and the final enactment presentations. In addition to data collection during workshops, participants were also asked to fill out a post-workshop questionnaire and write a reflection report after workshops. The post-workshop questionnaire required participants to describe their group's concept and its connection to the themes discussed in the affinity diagramming session, as well as their personal opinions on important themes, and reflections on the positive and negative aspects of concepts and themes. In the reflection report, we asked participants to reflect on the activities and learnings from workshops. The collected data is listed in Fig. 1.

3.6 Analysis

We employed reflexive thematic analysis (RTA), following the guidelines of Braun and Clarke [11, 12]. RTA involves iterative steps of identifying concepts and constructing overarching themes, emphasizing the researcher's active engagement and reflective involvement with the data [12, 20]. Aligning with similar approaches used in HCI studies with design workshop data [18, 65], our interpretation of RTA was adapted slightly to account for both textual and visual data, and aimed not only to construct meanings, but also to identify design opportunities. Below, we detail our analysis process based on the steps outlined by Braun and Clarke [11]:

Familiarization. After the workshops, the first three authors gathered to note the facilitators' observations. Then, the first author thoroughly reviewed video and audio recordings of the brainstorming sessions, affinity diagramming discussions, and enactment presentations, transcribing and noting down important parts where participants mentioned augmentation technology or transhuman abilities. Using this, the first author created a map of workshop phases and data sources. The authors examined the map, with the first author referencing data excerpts and notes from each phase and source. They agreed that the first author would lead the analysis, with the second and third authors acting as a sounding board. They also decided to use transcriptions, notes, and responses from questionnaires and reports. Idea presentation recordings were omitted since they were discussed during the affinity diagramming process. Instead, the first author referred to photos of ideas and diagrams when necessary.

Initial Coding. The 1st author coded the data inductively and iteratively, refining the set of codes in a second round. 29 codes were created (10 in Sensory, 6 in Physical, 5 in Cognitive and 8 in Emotional augmentation workshops), identifying different ideas that arose throughout the workshops.

Searching for and Reviewing Themes. The 1st and 3rd authors (both design researchers) held two meetings to discuss and develop initial themes from the ideas generated. Each code and associated data excerpt was reviewed to ensure they depicted or related to communication practices. All ideas were considered at this stage, and initial themes were formed by grouping and building connections between similar ideas based on augmentation types. For example, ideas like "network of minds" and "connected clones" focused on establishing connections among transhumans, with varying degrees of permanence. Authors created separate themes for permanent and temporary connections, but grouped them together under a single cluster. Following this approach, individual HAT themes were formulated and grouped into thematic clusters.

Reporting. The 1st author drafted the HAT themes and clusters and shared them with the other authors. All authors collectively finalized the themes and clusters. The research team also contributed critical perspectives. The second author brought insights from feminist, queer, and disability studies, while the fourth author contributed from translation studies and semiotics. Drawing on design research experience, the first and third authors focused on formulating HAT speculations that could guide research into communication experiences using current and near-future technologies. After initial themes and clusters were formed, the first author revisited the data based on reviewers' comments, identifying additional quotes from the participants on the potential impact of ideas on communication experiences and society as a whole.

4 Findings: Speculations for Transhuman Communication

In this section, we present, 8 SDs for HATs formulated on the workshop data and grouped under 4 distinct clusters (Fig.4). We present each HAT by (1) first summarizing the core idea behind the

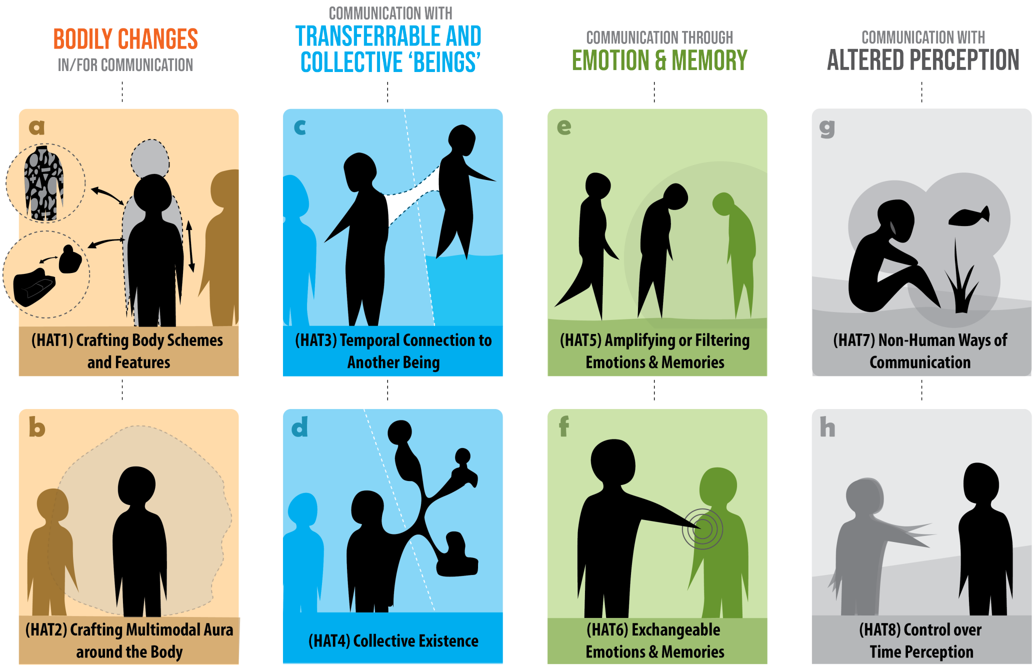

Fig. 4. An overview of human augmentation technology (HAT) speculations for transhuman communication

HAT speculation, then (2) exemplifying the different interpretations of the HAT with example ideas articulated throughout the workshops, and (3) reporting the reflections of the participants about the utilization of HATs in communication experiences and broader society. Throughout the text, we use *Physical*, *Sensory*, *Cognitive* and *Emotional* to refer to the sources of the data from workshops with regard to the respective themes.

4.1 Cluster 1: Bodily Changes in/for Communication

Our first cluster showcases HATs that give transhumans the ability to influence their physical existence of their body, changing it and reversing the changes at any time.

4.1.1 HAT 1: Crafting the Body Scheme and Features. The first HAT speculates abilities for transhumans to change their bodily features for communicative purposes (Fig. 4-a). Example ideas were transhumans *changing their body height* (*Sensory*), *skin colors and textures* (*Physical* and *Emotional*), as well as having *flying body parts* (*Physical*). The former was illustrated with a scenario where a transhuman adjusted their height to adapt to their conversation partner: *"Oh, let me come to your level! [transhuman shrinking down to adjust]"* (*Sensory*). Skin changes were perceived as an opportunity to express emotions (*Physical*) and to customize emotional expressions: *"My happiness can be pink and yours could be textured"* (*Emotional*). The concept of detachable and flying body parts was also mentioned as a possibility for the same purpose: *"If I am excited while talking to you and my head starts to detach from my body and raise up"* (*Physical*). By taking body modifications to the extreme, the participants also speculated that transhumans could *transform into things* and, by doing so, use the things' features as new vocabularies for communicative purposes. For instance, participants considered transforming into a soft couch to *"give a nice and cozy experience to people"*, or being a house for *"sheltering and protecting people"* (*Sensory*).

Reflections on these transhuman abilities brought up several issues: One topic of discussion was about the *impact of the ability to craft their body on social conventions*: Participants highlighted that the ability to change body features could deepen the *current normative perspectives: e.g., "looking young forever"* with skin change (*Emotional*). On the other hand, the discussions suggested *new social conventions* could be negotiated. For example, social norms and what is trendy might be redefined as new body alterations emerge: *"Like back in 2090 they were all with patterns, but now we are too busy so we are all monochrome now"* (*Physical*). Here another participant exemplified a potential norm by focusing on a funeral context: *"If you go to funerals, you might need to look a certain way, like you might not be allowed to be so colorful or be colorful depending on the context"* (*Physical*). In a similar vein, one participant stated "*... I could imagine it might be rude if I don't adjust [your height] to the level [of the conversation partner].*" On another note, while changing body schemes and features was considered a problem since these changes *"could make it hard to recognize people"* (*Physical*), it was also perceived as an opportunity to come up with *new encryption methods (i.e., fuzzing parts of the body)* to preserve the privacy of individuals against surveillance technologies (*Physical*). Finally, regarding body alteration, another issue discussed was related to the *autonomy of transhumans in initiating these transformations*. It was acknowledged that it might be challenging to conceal bodily changes if they occur unconsciously: *"For instance, becoming red when you are angry, or small when you are shy... would help to express feelings, but it would be difficult to hide them in case we don't want to show them"* (*Physical*).

4.1.2 HAT 2: Crafting a Multimodal Aura around the Body. Extending the bodily changes to the space around the body, the second speculative HAT focuses on the ability to create multimodal auras (Fig. 4-b). The utilization of the auras in conversations is exemplified in scenarios such as *distributing scents* that echo the conversation partner's *"mother's cooking"* (*Sensory*) or *"sprinkling some pheromones to be flirty"* (*Sensory*). Furthermore, participants highlighted the possibility of adjusting one's expression contextually with, for example, *an emotional screen* - a field around the body that amplifies or blocks their emotional expression while crying (*Emotional*). Apart from visual, olfactory and chemical auras, another idea was hinted at in the enactment presentation of the *physical* augmentation workshop by illustrating transhumans *using vibration signals* to communicate with each other underwater.

As demonstrated in these ideas, while auras were considered for enriching communication experiences via *non-verbal cues*, one issue discussed was that auras could be utilized for *social deception and manipulation*, i.e., hiding stigmatized behaviors such as crying in public (*Emotional*) or body odors such as sweat (*Sensory*). By citing the strength of the sense of smell in recalling memories and indicating changing moods, one participant mentioned that scents might be utilized for manipulating individuals (*Sensory*).

4.2 Cluster 2: Communication with Transferable and Collective 'Beings'

The second cluster presents speculative HATs in which transhumans are not bound to a singular physical body: Instead, participants designed flexible, transferable, ephemeral and collective ways of being and of having connected bodies.

4.2.1 HAT 3: Temporal Connection to Another Being. This speculative HAT describes how transhumans can temporarily share their bodies with other transhumans (Fig. 4b). This HAT was illustrated with scenarios where two beings *share the control and the perception of one body* by using one of their bodies as a host. For example, one enactment scenario included a couple sharing an intimate camping experience with a remote partner being transferred to their partner's body (*Cognitive*,

Fig. 3c). The remote partner was connected to this experience sitting at home, but having access to the host's perception and body control with the host's consent. During this shared experience, the remote partner noticed an interesting plant. To communicate the interest, the remote partner moved the host body's arm toward the plant, depicting *"controlling a remote body" as a channel* for communication with the host. Moreover, regarding the risk of the remote partner focusing on the shared experience rather than paying attention to what happens in their physical existence, during earlier discussions about a similar temporal connection concept, participants highlighted a *"[body-embedded] processor that could record and process everything happening around you"*, so that the remote body would be notified in cases such as emergencies (*Physical*).

Participants' reflections on the temporal connection anticipated *a deeper empathy* and *resolution of conflicts* that might be achieved by providing a literal way of being in someone else's shoes: *"you can fully see something from their perspective. . . this could be really beneficial for humanity ...lots of conflicts, big and small ones, seemed to be because of lack of [such an] understanding"* (*Cognitive*). On the other hand, the participants acknowledged the risks of remote partner invading the autonomy of host bodies and suggested there should be *strict mechanisms for consent* and an *off-switch* for disconnection whenever the host wants to disengage (*Cognitive*).

4.2.2 HAT 4: Collective Existence. This speculative HAT explores how transhumans could be in a state of constant connection to each other, including a fine-tuned awareness of their mental, physical, or emotional states (and beyond)- see Fig. 4d. Related to this, in the *Sensory* augmentation workshop, mycelium networks were discussed as a metaphor for what a state of "collective being" for transhumans may look like. Mycelium networks are big fiber structures of mushrooms underground that allow communication among different parts of the network, as well as with plants at a molecular level [83]. Influenced by the mushrooms, this HAT considers a set of bodies that belong to the same consciousness, but their existence is not in one particular place, but in many. Interpretations of this idea also included *a network of minds* (*Cognitive*), e.g., brain chips allowing constant access to other transhumans' knowledge and perception, *connected clones* (*Emotional*), and *having multiple instances* (*Sensory*), e.g., the ability to create connected copies of a transhuman existing in different places but in a constant connection.

Reflecting on ideas around this HAT, participants discussed the potential of *conflicting emotions coming from different parts of the collective*: *"There are so many of me [a clone in a collective]. But then I am perceiving all the different emotions coming from all the copies of myself. I'll need to know how I am actually feeling"* (*Emotional*). Being connected to different entities was also considered a *potential for distractions: "In the future, it might be that you stand with each other, but you are completely somewhere else"* (*Sensory*). Finally, the ability to exist with multiple bodies evoked discussions about *ethical issues* regarding the relationships among bodies of a collective. For instance, participants discussed whether it would be okay to create an instance for yourself to date or to eat (*Sensory*).

4.3 Cluster 3: Communication Through Emotion and Memory

The third cluster presents two HAT speculations about transhumans having control over their own and other transhumans' emotions and memories.

4.3.1 HAT 5: Amplifying or Filtering Emotions & Memories. This HAT speculates on abilities to filter or amplify emotions and memories (Fig. 4e). Participants discussed multiple ways that emotions and memories can be modified: The first branch of the idea focused on *transhumans having the ability to modify their own emotions and memories.* For this, a funeral context was discussed as an example where a transhuman amplifies their own level of sadness to empathize with others (*Emotional*). The second line of ideas was about *altering the emotions and memories of others.* One interpretation of this idea was having *places that filter emotions and memories of crowds.* This idea was illustrated for a collective entertainment scenario, featuring a concert hall where transhumans' emotions are filtered depending on the mood of the song (*Emotional*). Another example was an ideation room where the memories related to a particular topic are suppressed to enable judgment-free discussions about the topic (*Cognitive*). This idea was motivated by discussing that preconceived notions tied to memories can sometimes impede open-mindedness about the topic. Similar to places, two other ideas from the *Emotional* augmentation workshop illustrated how *objects* and *auras around transhumans can be used to impact the other transhumans' emotions nearby*. For the former, one concept was an *emotional bomb* that modified the emotions of transhumans in a particular space when dropped. The latter was illustrated with another enactment presentation about a group of transhuman friends throwing a goodbye party for a friend, and a transhuman party guest who had an emotional aura affecting those around them.

The idea of amplifying/filtering emotions and memories illustrated positive outcomes such as *bringing transhumans together by enabling shared emotions in a particular context* as illustrated in the contexts of concerts and funerals. However, the scenario about the emotional bomb emphasized the importance of transhumans *retaining autonomy over their own emotions and memories* by illustrating how forcing emotions and memories might utilized for authoritarian purposes- in the enactment presentation, this bomb was used to oppress the anger of a protestor by the authority. Moreover, the enactment about emotional aura highlighted a similar autonomy issue with the person with aura feeling overwhelmed and ashamed by the sadness their presence unwillingly intensified in others. Furthermore, abilities to remove emotions and memories were considered as problematic by the participants, raising concerns regarding their role in decision-making: *"How do we make informed decisions if all the memories are removed?"* (*Cognitive*).

4.3.2 HAT 6: Exchangeable Emotions & Memories. This HAT presents new abilities for transhumans to record and transfer emotions and memories between each other (Fig. 4f). In this direction, one line of ideas was based on having *physical tokens to exchange emotions and memories*. In an enactment presentation of an illegal gambling scenario (*Cognitive*), transhumans extracted memories and emotions into physical tokens and used them instead of money. A similar token concept was illustrated in another enactment where a company produces emotion tokens that are sold to transhumans for them to plug into their body and feel emotions such as happiness (*Emotional*). In addition to tokens, participants also considered *synthetic skins that allow transhumans to exchange memories*: For instance, one enactment presentation scenario depicted two transhumans secretly exchanging a multisensory memory of a concert by touching their palms together (*Cognitive*, see Fig.3b). Moreover, speculations about exchanging memories and emotions brought up *the fidelity of emotions and memories* as an interesting aspect of communication. Illustrating this, in one of the enactment presentations (*Emotional*), intense happiness was stolen from a crowd and replaced with less intense happiness.

The participants discussed how this HAT could improve *empathy among transhumans* and *make rare emotions accessible to wider crowds*. For the former, they used an example of transmitting a crime victim's emotions to the perpetrator, helping them to understand the impact of their actions firsthand (*Emotional*). An example of rare emotions that can be more accessible through the exchanging of emotions was an *"emotion of a monk living in a cave for years"* (*Emotional*). On the other hand, there were also some concerns about commodifying emotions and memories as exchangeable notions. Reflecting on the idea of exchanging multisensory memories of a concert, participants noted potential challenges such as *copyright issues* and *privacy concerns* that these issues might be more critical when the whole sensory aspects of memories are captured and shared (*Cognitive*). Moreover, *identity loss* was mentioned as another potential consequence of exchanging memories when these memories were extracted and removed from individuals. This was illustrated within the gambling scenario (*Cognitive*) in which a transhuman attempted to gamble all of their memories by extracting them into physical tokens. A police authority stopped this but they used another token to simulate a devastating state of erased memories for the gambler. Finally, the enactment about company-produced emotion tokens (*Emotional*) demonstrated the *addictive potential of emotion tokens*: In this enactment, when the supplier ran out of happiness tokens, the transhumans suffered as they lost the ability to feel happy without the tokens.

4.4 Cluster 4: Communication with Augmented/Altered Senses and Perception

This cluster includes speculations for HATs regarding altering or augmenting the ways that transhumans sense and perceive the world.

4.4.1 HAT 7: Non-human Ways of Communication. This speculation considers transhumans developing augmentations to bridge the ways humans and non-human animals or plants communicate (Fig. 4g). The ideas around this HAT included *imitations of non-human communication channels* such as sending and receiving vibrations to communicate and using skin augmentations to photosynthesize. For example, one enactment presentation (*Physical*) portrayed a future where Earth's land becomes uninhabitable, prompting some humans to live underwater. To adapt to the new environment, they integrated a gill-like technology into their body to breathe underwater and *communicate through vibrations (as some fish do) and signaling with lights (similar to Morse code).* In a similar vein, although it was not speculated as a direct channel of communication, *skin augmentations to photosynthesize* were also mentioned as an intimate way of sharing spaces with plants: *"The potential uses of [photosynthesis] would be similar to sunbathing together that would represent intimacy"*. Finally, in the *Sensory* augmentation workshop, the participants considered a non-detailed technology that might allow humans to speak verbally with non-human animals such as dogs and tigers.

One central issue about communication with animals and plants was about the tension between *feeling more connected to non-humans* and their *potential exploitation* with these newly established abilities. For example, on one hand, participants highlighted that photosynthesizing together with plants might make transhumans *"more connected to nature as well, like the kind of oneness of everything"* (Sensory), yet, on the other hand, they also mentioned that *" [...] there could be class divisions based on access to the resources for photosynthesis"* (Sensory). This tension was also captured in the enactment with transhumans that can communicate underwater (Physical), and while sharing vibration and light as a common channel of communication enabled collaboration between land and underwater transhumans (in this case to solve a crime together), it was exploited by a land transhuman to deceive and kill the underwater ones at the end.

4.4.2 HAT 8: Control over Time Perception. This speculation is based on a transhuman ability to slow down or speed up the time perception for communication (*Cognitive*, Fig. 4h).

Reflecting on this idea, participants foresaw that slowing down time would give individuals *time to process and immerse themselves in communication experiences*: *"[manipulating time perception] could have a range of applications such as in education or entertainment. In education, by allowing students to slow down the pace of lectures or demonstrations to better understand and absorb the information being presented, and in entertainment, by allowing people to experience fast-paced events or actions in slow motion, which could enhance the viewing experience and make it more immersive."* (*Cognitive*). Furthermore, a participant mentioned that this technology could allow people (i.e., loved ones) to agree on a specific time to slow down their perception, so as to *experience more in a limited time.* On the other hand, one critical issue discussed regarding this HAT was that it might be *impolite* to speed up time when in conversation with others: *"After you have a long discussion* *with someone and you ask a question. If they don't react, you might realize that they have speeded up the time."* (*Cognitive*)

5 Discussion

Our work reveals design opportunities for human augmentation technologies for developing and researching the near future of our communication. In what follows, we discuss potential design opportunities and challenges for transhuman communication technologies based on the insights gained through the speculative endeavors with participants. In the last subsection, we point out how these issues can already be researched by crafting state-of-the-art technologies. We introduce these ideas as starting points and as inspiration for future conversations about transhumanism and communication in CSCW and beyond. It should be noted that we do not understand these concepts in absolute, concluded terms, but more as potential avenues for further researching communication experiences.

5.1 Modes for Sign Production in Transhuman Communication

The HATs highlight the potential of new *transhuman modes of sign production*: Cluster 1 (crafting bodily temporalities in/for communication) augments existing communicative strategies such as body language, through transhumans changing height or transforming into non-living beings. Within the same cluster, participants' speculations also mimicked other semiotic systems by imagining transhumans changing their skin color or emitting smell to express their feelings, in a similar way we wear clothing or perfumes. These speculations echo explorations within current research, e.g. prosthetic tails [106] or wearable technologies that change color [55], or dispense smell [2, 52]. However, participants' speculations also went beyond current ideas: HAT 2 (crafting multimodal auras), HAT6 (exchangeable emotions & memories) and HAT 7 (non-human ways of communication) propose signs that are no longer solely visual or olfactory. Instead, they envision transhumans producing signs across all senses that influence other transhumans on a fundamental level to evoke specific feelings, manipulate them, or to communicate through chemicals as some plants and non-human animals do.

In this regard, our work reveals opportunities for *new modes of sigh production [27]* to increase *semiotic diversity*. As we mentioned in the background and related work section, meaning-making involves the author's intent, the reader's interpretation, and the medium's structure (as outlined by Eco [28]). From this perspective, these new transhuman modes of communication propose new structural properties that are arguably more ambiguous than conventional modes of gestural or verbal communication. For instance, imagine an awareness event on the disruptive impact of colonizing planets, where transhumans with abilities to evoke emotions through multimodal auras (HAT 2) and exchange emotions (HAT 6) are present. In a heated discussion between two transhumans, one evokes excitement while opposing the other. The interpretation of this action raises questions: Does it indicate enjoyment or does it aim at intensifying the debate? Similarly, controlling someone else's body in communication (as exemplified in HAT 3 by a connected transhuman moving the host body's arm during a climbing trip) can lead to varied interpretations based on the context and specifics of the control. These examples highlight that these modes of communication are open to interpretation, influenced by factors such as the relationship between transhumans, the form of the message, and the intentions of senders and receivers. The ambiguity of these forms of communication offers the potential to diversify communication experiences in ways that might not be possible to imagine with our current abilities and communication practices. By revealing these forms, we believe that our work paves the way for CSCW and HCI researchers to study what kinds of sign production and meaning-making processes these new modalities might yield in social experiences.

5.1.1 Challenge: Achieving Common Ground. On the other hand, these new modes of sign production might intensify the challenge of finding common ground [19] in communication. Common Ground theory [19] suggests that individuals engaged in conversation must share knowledge in order to be understood and have a meaningful conversation. While the issue of common ground is not unique to transhuman communication and has been a subject of CSCW research as an already-apparent challenge in our society (e.g., finding common ground among neurodiverse dyads [108] and different cultures [31]), our work highlights that these challenges might intensify with HATs. For example, our HATs suggest that not having special communicative abilities might deepen costs for finding common ground, maybe even to the extent of causing a technology-divide within societies. Without common ground in the abilities that the conversation partner have, the individuals might be open to manipulation and deception (as exemplified via distributing scents in HAT 2). Moreover, the new cultures and social etiquette around these specific abilities (i.e., not adjusting your height to a conversation partner being seen as an impolite behavior: HAT 1) might add new barriers in personal communication experiences. On this note, the ability to customize physical appearance might make it hard to find common ground between individuals if individuals become unrecognizable with extreme changes. All in all, our work suggests that the designers of transhuman communication technologies should focus on making these technologies easier for people to find common ground when their abilities significantly vary and allow extreme modifications to make themselves unrecognizable (i.e., not having a recognizable face as seen in HAT 1).

5.1.2 Challenge: Relation to the Status-Quo. Our findings suggest that transhuman technologies to reconfigure physical existence could challenge the current norms, and the body/mind of the future could be crafted at will (i.e., transforming into a couch in HAT 1). On the other hand, HATs can lead in the opposite direction by confirming current norms, for instance with transhumans changing their skin to "look young forever" as highlighted in HAT 1. Indeed, our HATs might deepen the structural problems regarding discrimination based on judgments about how a person looks from a particular normative perspective [101]. Similar tendencies can be observed in Social VR studies: Although virtual spaces allow individuals to exist and express themselves freely (i.e., with cross-gender avatars [35] or avatars expressing emotions with body modification [6]), avatar creation choices of individuals in Social VR such as Second Life are highly influenced by gatekeepers' avatars reflecting the Western beauty standards of ideal body sizes and light-colored skins, and neglecting others [72]. Thus, although HATs might promise an insurmountable amount of ways for transhumans to physically express themselves, the designers of transhuman communication technologies also need to study the possible risks of extending the consolidation of the status-quo to other modalities (i.e., chemicals, emotions or memories that individuals express themselves through).

5.2 Collective Meaning-Making & Empathy in Transhuman Communication

Our findings suggest that HATs challenge boundaries between entities. By promoting a relational view of *being* and *communicating*, they blur the line between bodies and things. This is similar to the JIZAI body concept [48] where transhumans can clone themselves or fuse multiple bodies in one existence. Moreover, our findings resonate with emerging live streaming services, for example in terms of proposing technologies for constant or temporary connections among individuals [80, 104]. Yet, our work expands these by illustrating how HATs can enable a *deeper empathy* through extending the modalities of connections toward sharing sensations and emotions that can contribute to *collective meaning-making* processes. These are exemplified with the multimodal connections seen in Cluster 2, and transhumans sharing emotion in a concert in HAT 5. Our work also suggests that possibilities for empathy with non-humans can be extended when transhumans adopt modalities that non-humans such as plants use (HAT 7). We believe that these ideas complement communication complexity by bridging entities together and achieving a better understanding of how they perceive the world. For example, accessing temporal connections (HAT 3) can complicate meaning-making by acknowledging the host's perspective while incorporating the transferred being's interpretation and past experiences. Or, the introduction of a transhumans' ability to share collective experiences (HAT 4) opens up new opportunities for meaning-making by combining diverse experiences and feelings within the collective. For instance, if we imagine a scenario where a transhuman in a collective of animal lovers is exposed to a demanding dog on the street while they are in a rush - instead of getting angry immediately, they can interpret the situation by leveraging the experiences of why other parts of the collective love animals, and their feelings of anger might ease.

On the other hand, control over transhuman bodies (HAT 3) can also facilitate joyful engagements that can contribute to social cohesion. These can be similar to playful interactions highlighted in other research [14, 33, 76]. For example, Arm-A-Dine [70] features a network of prosthetic arms on people that are controlled through the affective responses of others, and exemplifies how controlling other bodies might lead to a joyful social experience. While the example of HAT 3 connecting to and controlling another's body provides the same direction, our HATs further extend such possibilities towards perceptions, emotions, and memories. To illustrate this, we can extend the idea of the "ideation room" (HAT 5) where certain memories of individuals are suppressed for judgment-free ideation. This might open up cases where transhumans voluntarily give this control to the room so as to forget memories about bottle usage, and communicate with each other to achieve new forms for containing water for mobile usage. Similar to this idea, consensual control over or accessing other transhumans' perceptions might open up competitive or collaborative play experiences. For instance, transhumans could speed up each others' time perception to prevent them from observing the environment in competitive gameplay or slow them down to enable collaboration. Furthermore, the concept of modifying time perception may challenge current norms, particularly for disabled individuals who face difficulties due to the prevailing "normative" perspective on time [59]. In a future where time perception is modifiable, the established norms regarding time could be challenged, encouraging a diverse range of time perceptions for communication experiences.

5.2.1 Challenge: Autonomy, Agency & Privacy. *Autonomy* appeared as one of the key topics in HAT speculations, with considerations arising from the modification of bodies, emotions, memories, and perceptions for communication purposes. For example, participants worried that involuntary changes in their bodies linked to emotions could be tough in social settings (HAT 1). These considerations align with existing discussions on bodily autonomy in the context of human-technology integration, where maintaining a sense of ownership over integrated technology is crucial [87]. Moreover, similar to Cluster 2 (Transferable and Collective 'Beings'), studies about live streaming suggest that constantly streaming, even during intimate relationships, evokes privacy and agency concerns in streamers as this gives access to things that they would prefer others not to see and hear [80]. While confirming these concerns for transhuman communication, our HATs propose an extension to state-of-the-art technologies, giving others access and control to the bodies, emotions and memories of transhumans. For instance, our HAT regarding modifying the emotions of others (HAT 5) illustrates that while applying this ability to a crowd can have a naive goal of uniting people in a concert setting, the same technology can be utilized for eugenic ideologies, taking the agency away from the individuals and adjusting them to fit predetermined standards [54].

5.2.2 Challenge: Reductionist Transhuman Communication. Another challenge in designing HATs for communication is developing them with *a reductionist view toward human communication.* Some of the HATs could be easily interpreted in this direction. For example, Cluster 2 (Transferable and Collective 'Beings') might lead to an understanding where communication can be perceived as the ability to possess different bodies to reach ubiquity, to teleport one's consciousness around the world, and to come to a perfect understanding of other entities. Semiotically speaking, this might lead to an assumption that an identity of the conscience is transmitted and can be accessed perfectly - and therefore suggesting a form of communication, but not one of signification, as the interpretation (that is, the way in which the body reads the different minds it hosts) is fixed as in digital communication. Transhumans can emerge from these ideas as beings of pure mind, whose bodies and feelings are commodified, interchangeable and ultimately disposable, as they do not meaningfully participate in the formation of the transhuman being. Instead of assuming that access to other beings diminishes the personal identities of individuals, designers of future communications technologies should pay attention to how HATs could contribute to the meaning-making processes. For example, instead of interpreting HATs for connected beings (Cluster 2) as bodies & minds without individual consciousness, researchers should focus on how individuals could negotiate and interpret meanings when they have access to each other sensations. Similarly, instead of considering emotions as perfectly exchangeable notions, researchers could focus on how transferring sensations (i.e., heart rate or body temperature changes) might be interpreted in communication.

5.3 Researching Transhuman Communication through State-of-the-Art Technologies

As highlighted at the beginning of the discussion section, rather than being conclusive, the points raised above are possibilities constructed based on our SD and RTA practices. To deepen our understanding of these discussion points to some degree, we can turn to existing technologies and use them as prototypes to illustrate possibilities [56] in speculative enactments [32]. Speculative enactments are designed speculative contexts with artifacts with which participants can engage and enact, and which allow empirical analysis over enacted experiences. Investigating near future technologies through the lens of HATs in such enactments could yield a deeper understanding of transhuman communication experiences before they become a reality. In what follows, we highlight how some of our findings might be used for future research.

To explore Cluster 2 (Communication with Transferable and Collective 'Beings'), wearable cameras and microphones (similar to body-worn live-streaming devices [80]) can be explored and extended with different modalities to access the senses of others. These can be complemented by creating real-time awareness of biosignals between entities through vibration or visual feedback [30], and electrical muscle stimulation (EMS) [94] can be used to achieve remote control of individuals. By using similar setups, researchers can already simulate experiences related to Cluster 2 in experimental conditions, which would help to understand and explore these HATs in action. For example, researchers could create prototypes with different levels of agency (i.e., full control and limited control over the host body) and access to different bodily information and senses (i.e., biometrics, vision, smell, haptics). User studies can be designed to identify individuals' perspectives toward issues of privacy and autonomy, or opportunities regarding empathy. Identical systems can also be created to record and exchange multisensory memories, and a longitudinal study could be conducted with (e.g., with couples) to understand the reasons individuals share multisensory memories with each other. Similarly, HAT 7 can be explored through wearables with sensors that translate the intensity of chemicals or micro-vibrations released by plants. Conducting studies with these wearables can focus on how such systems help them attune [5] and feel connected to non-humans.

Cluster 1 can also be explored through wearables, as well as VR environments to imitate extreme body modifications and multimodal visual and olfactory auras during communication scenarios, similar to the study by Bernal and Maes [6]. For instance, changing height can be achieved by height-adjustable shoes [73, 89]. In-situ studies with these prototypes can be done in contexts such as dating, teaching, or night events in order to understand the ways individuals use body modifications in communication. Also, Social VR applications can be created based on the abilities featured in Cluster 1, and behaviors and perceptions of individuals can be studied to better understand the risks and opportunities around social norms and the status quo. VR environments can also be a resource for imitating time perception change (HAT 8) that could be studied in terms of its impact on social interactions (i.e., how conversation partners react when someone speeds up the time.)

Finally, for experiences such as collective feeling and exchanging emotions (Cluster 3), the Collective Effervescence of Durkheim [88] and Interaction Ritual Chains by Collins offer related societal concepts [22]. These notions suggest that the social atmosphere is created by the collective actions of individuals, be it in simple daily interactions such as a greeting or a mass event such as a concert. A previous study examining interaction ritual chains in streamed Twitch concerts found that large-scale interaction ritual chains did not work successfully since physical factors such as feeling a body next to you do not exist in the virtual environment [97]. Here, we believe that an interpretation of sharing emotions or an exchange of bodily arousal states might contribute to a shared sense of spaces for interaction rituals. To explore this, for example, wearables can be used to measure the biometric data of participants (i.e., skin conductance) during a concert experience and then exchanged among participants in real-time via vibration or visual feedback [79]. Another possibility here might be to design interactive ingestible prototypes (similar to the ingestible camera proposed by [33]) that can be controlled through devices such as smartphones, to emit chemicals that influence how participants feel. User studies with such implementations could reveal a deeper understanding of the possibilities for empathy, or challenges regarding autonomy and privacy about the issues raised in HAT 6.

6 Limitations

While our work explores diverse forms of communication through HATs, it is important to acknowledge the limitations of our findings, which are influenced by the backgrounds and perspectives of the workshop participants. The predominant STEM background of the participants and their gender being men (despite our efforts to balance the ratio) may have restricted the exploration of social implications related to HATs representing the wider society. To address this, we incorporated activities like body-storming [67] and enactments to encourage experiential and critical perspectives by role-playing different characters in speculated scenarios. These activities helped participants to enact characters overarching current gender & human-technology classifications (i.e., underwater transhumans in HAT 7), aligned with Haraway's criticism of such dichotomies [41], and revealed potential issues such as autonomy, exemplified by the concept of using an "emotion bomb" to calm people (HAT 5). However, we acknowledge that our methodology (i.e., co-speculation workshops and thematic analysis) let us capture only a subset of possibilities and issues that were found to be relevant by our workshop participants and the authors. In this regard, our work did not focus on evaluating whether our findings captured the most major issues (or whether it missed any major issues). In future work, including participants from social science disciplines or art backgrounds and conducting similar studies could further diversify speculations and provide additional insights. Furthermore, although our work did not aim to quantitatively evaluate the perception of individuals about the (societal) impacts and/or relative urgency of transhuman communication technologies, by presenting these first speculations on HATs for transhuman communication and some potential issues regarding them, our work lays a foundation for future work that could identify the perceptions of wider society through quantitative studies (e.g., online surveys over the issues and speculative HATs outlined in this study).

7 Conclusion

This paper presents 8 design speculations for human augmentation technologies to be used for communicative purposes, thematically grouped under 4 clusters: (1) Bodily Changes in/for Communication, (2) Communication with Transferrable and Collective Beings, (3) Communication Through Emotion and Memory and (4) Communication with Augmented/Altered Perception. These are formulated based on a series of co-speculation workshops on transhuman communication, and through a reflexive analysis of the workshop data. The HATs presented in our work introduce new possibilities for diverse and nuanced forms of communication. They offer innovative modes of sign production, such as evoking emotions in others and controlling their bodies. Additionally, these HATs enable collective meaning-making practices by connecting to and existing as a collective with other beings. On the other hand, the study highlights how such extensions of our communication modes can raise issues such as challenges in finding common ground in conversations, or deepening the structural problems of our society when individuals exist with diverse sets of abilities, new cultures, or fluid physical existences. Moreover, our exploration suggests that abilities to establish connections among individuals and to exchange, modify or control others might intensify concerns regarding privacy, autonomy and agency, as well as pave the way for developing reductionist communication technologies that pose our physical and individual existences as commodities. By highlighting these opportunities and critical issues, our work contributes to a design space that delves into the intricate and diverse realm of transhuman communication.

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